Data 227 - Autumn 2026 HW 3

Data Visualization and Communication - Trimble

Due: Thursday October 22, 2026 11:59pm

Fireball data

1. The Center for Near Earth Object Studies has collected a database of large, bright objects that have impacted the Earth and were detected by the flashes they made, day or night, in the atmosphere (Fireball and Bolide Data, Center for Near Earth Object Studies). As of Oct 10, 2026 the list has 888 events with latitudes.

   `https://cneos.jpl.nasa.gov/fireballs/`

   This question asks you to create effective visualizations of the distribution of variables related to these fireballs.

   a For all of the columns in the database, consider the nature of each variable–should we expect the variables to be numerical (discrete or continuous)? Categorical (nominal or ordinal)? Write down what you expect without viewing the storage types of the data and compare to the storage types.

   b Read in the data and convert the data to suitable types. Convert the date and time to a time data type.

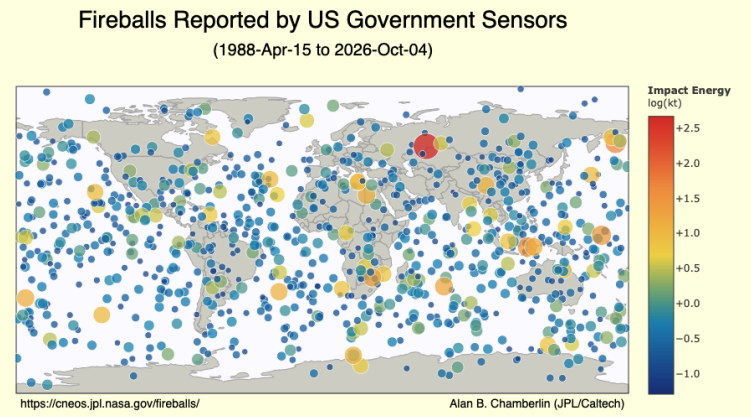
   c Is there anything unusual about the distribution of date and times of fireballs in the dataset? Are there places in the dataset (years, months, latitudes) that get more / fewer impacts?

   d For visualizing distributions, histograms or cumulative distribution plots are suitable. Visualize the distribution of longitudes using a histogram, and describe the center, shape, and spread of the distribution. Indicate the presence of any outliers. Is this distribution consistent with what you might expect?

   e Make a Q-Q plot or an equivalent plot of the distribution of the latitudes of fireballs relative to the theoretical distribution that fireballs occur at a density proportional to the cosine of latitude.

   f For right-skewed graphs, a log-scale is often appropriate for visualizing differences in the data. Plot the distribution of Total Impact Energies using a histogram, and describe the center, shape, and spread of the distribution. Indicate the presence of any outliers. (You may be able to plot log-bins in plotting software, or you may need to create them yourself.)

   g Count the number of fireballs in each degree of longitude. Make a plot that compares the histogram-of-the-histogram (number of degrees of longitude with 0, 1, 2, etc. events) with the Poisson distribution with the same number of average events. This will be a Poisson distribution with a mean of 888 / 180. Comment on their agreement / disagreement. (2pts)



Open-ended Weather visualization

2. Get some data on the weather in Chicago.

   NOAA has historical weather at thousands of weather stations https://www.ncei.noaa.gov/access/past-weather/A https://www.ncei.noaa.gov/cdo-web/search. Third-party commercial weather-marketing companies might also serve you data, but you'll probably need to sign up for an API key if you want tens of thousands of rows.

   Make two visualizations that summarizes some fact about the past 30 years of weather in Chicago. (Clearly, each possible number representing annual weather has 30 values over the time since 1996.) Questions that pop to mind are: At what date is the first frost? Which months are most likely to include a basement-flooding downpour? Is it cloudier in spring and fall, or does it just seem that way?

   Make one visualization that reports some fact about hourly weather that uses at least 5 years of weather data. You can highlight unusual events (rapid changes in temperature), disaggregate events like precipitation, temperature, fog, hail, cloud cover... by time of day, or count heating degree-days to compare to your gas bill.

   There are many examples of weather visualizations that can be found on the internet; cumulative rainfall by date, high and low temperatures over the decade by date, maybe even cloud cover categories by date or time of day. Try to show off.

